Smart Hostel & Accommodation Management System (VISTA)

Functional Testing – Test Cases (with Pass and Fail Examples)

Purpose Functional testing verifies that VISTA works as specified in the functional requirements (FR01–FR35) of the Lab VI report: authentication and roles, complaints and escalation, visitor approval, hostel services, SOS, peer-to-peer delivery with OTP/QR handover, notices, community features and administration.

Report date: 18 Oct 2026, 22:27 IST · Results source: test-results/test-results.json (functional) · Environment: Linux 6.18.44-fc-v139, 2 vCPU, Python 3.13.16, Node v22.22.0, PostgreSQL 16.15 (Ubuntu 16.15-0ubuntu0.24.04.1), Chromium 141.0.7390.37

Pass executed, met the criterion | **Fail** executed, did not meet the criterion | **Blocked** could not run in this environment | **Not Run** needs resources not available

TC ID	Test Case Description	Module	Pre-Conditions	Test Steps	Test Data (Input)	Expected Result	Actual Result	Status	Defect ID
FT-01	**Login with valid student credentials**	Authentication	Seeded student account exists	1. Open login page 2. Enter email and password 3. Click Sign in	priya.sharma@vista.demo / Student@123	Token issued; student dashboard opens with role 'Student' and SOS button	API: HTTP 200; JWT issued; role=STUDENT; room=G-101 UI: redirected to dashboard 'Good evening, Priya', role label 'Student', SOS button visible	Pass	—
FT-02	**Login with an invalid password**	Authentication	Account exists	1. Enter valid email + wrong password 2. Sign in 3. Repeat with unknown email	arjun.nair@vista.demo / NotMyPass1	401 'Invalid email or password'; same message for unknown email; stays on login page	API: HTTP 401 'Invalid email or password' for wrong password and for unknown email (same message) UI: error 'Invalid email or password' displayed; URL stays /login	Pass	—
FT-03	**Role-based dashboard and menu**	Authentication	Student and warden accounts	1. Log in as student 2. Log in as warden 3. Compare dashboard cards and menu	Priya (student), warden.lh	Student sees student cards + Vault; warden sees management cards (SOS, escalations) + Users	API: Student dashboard cards=['active_deliveries', 'active_services', 'open_complaints', 'pending_visitors']; warden dashboard has 8 management cards incl. active_sos UI: student menu shows Document Vault, no Users; warden menu shows Users/Hostels, no Vault; warden dashboard shows Active SOS card	Pass	—
FT-04	**Student attempts administrator operations**	Access control	Logged in as student	1. Call admin user creation, audit log, visitor decision, notice publishing 2. Open /admin/users in UI	Student token	Backend returns 403 for every call; UI shows 'Access denied'	API: Student received 403 Forbidden on POST /api/admin/users, GET /api/admin/audit-logs, PATCH /api/visitors/1/decision, POST /api/notices UI: /admin/users as student shows '403 Access denied' page	Pass	—
FT-05	**Student self-registration with valid data**	Registration	Email and reg. no. not used	1. Submit registration form 2. Submit again with same reg. no. 3. Log in with new account	25BCE9001, new.student@vista.demo	201 Created as STUDENT; duplicate reg. no. -> 409; new account can log in	API: 201 Created (role STUDENT); duplicate reg no -> 409 'An account with this registration number already exists'; new account can log in	Pass	—
FT-06	**Registration input validation**	Registration	None	1. Submit form with invalid reg. no., phone, weak password, bad email, script in name	ABC123, 12345, 'short', '<script>'	Each invalid field rejected (422 / inline error); nothing saved	API: reg_no='ABC123' -> 422 'reg_no: Registration number must look like 24BYB0193'; phone='12345' -> 422 'phone: Phone must be a valid 10-digit Indian mobile number'; password='short' -> 422 'password: Password must be at least 8 characters and contain letters a'; email='not-an-email' -> 422 'email: value is not a valid email address: An email address must have '; full_name='<script>' -> 422 'full_name: Name may contain only letters, spaces, dots, hyphens and ap' UI: inline errors for reg. no., mobile, password and name; form not submitted	Pass	DEF-07 (fixed)

TC ID	Test Case Description	Module	Pre-Conditions	Test Steps	Test Data (Input)	Expected Result	Actual Result	Status	Defect ID
FT-07	**Update profile and roommate preferences**	Profile	Logged in as student	1. Edit mobile + preferences 2. Save 3. Reload page 4. Try invalid phone	9988776655, QUIET / LATE	Changes saved and shown after reload; invalid phone rejected	API: Profile saved and re-read: phone=9000011111, pref_sleep=LATE; invalid phone -> 422 UI: 'Profile updated' toast; after reload mobile=9988776655, study=QUIET	**Pass**	—
FT-08	**Register an electrical/plumbing complaint**	Complaints	Student with room	1. New complaint 2. Choose category, title, description 3. Submit	PLUMBING, 'Bathroom tap leaking'	Complaint code CMP-xxxxx created, status OPEN, forwarded to maintenance staff who is notified	API: 201; CMP-00008 OPEN, priority MEDIUM; auto-forwarded to Ramesh Electricals (MAINTENANCE) and notified UI: CMP-00008 created (status Open, forwarded to maintenance staff) and still listed after page reload	**Pass**	DEF-07 (fixed)
FT-09	**Complaint with missing required fields**	Complaints	Logged in as student	1. Submit empty complaint form 2. Submit without category 3. Invalid category	Empty title / description	Inline errors in UI; API returns 422; nothing stored	API: Empty title/description -> 422 'title: String should have at least 5 characters'; missing category -> 422; invalid category -> 422 UI: three inline errors shown (category, title, description); dialog stays open, nothing submitted	**Pass**	—
FT-10	**Washroom and water complaints get HIGH priority**	Complaints	Logged in as student	1. Raise WASHROOM complaint with LOW priority 2. Raise WATER complaint with MEDIUM	WASHROOM/LOW, WATER/MEDIUM	Both stored with priority HIGH	API: WASHROOM requested LOW -> stored HIGH; WATER requested MEDIUM -> stored HIGH	**Pass**	—
FT-11	**Staff updates complaint status to resolved**	Complaints	Complaint assigned to staff	1. Staff marks IN_PROGRESS with note 2. Marks RESOLVED 3. Student closes	Notes: 'Visiting at 4 PM', 'Latch replaced'	Status changes saved, history shows each step, student notified	API: OPEN -> IN_PROGRESS -> RESOLVED by assigned staff, CLOSED by student; history has 4 entries; student notified	**Pass**	—
FT-12	**Invalid complaint status changes are blocked**	Complaints	Open complaint	1. Staff tries CLOSED 2. Student tries RESOLVED 3. Another student opens it 4. Change a CLOSED complaint	-	403 / 404 / 409 respectively; status unchanged	API: Staff CLOSE -> 403; student RESOLVE -> 403; other student view -> 404; CLOSED -> IN_PROGRESS -> 409 'Invalid status change from CLOSED to IN_PROGRESS'	**Pass**	—
FT-13	**Escalation of unresolved complaint to chief warden**	Complaints	HIGH complaint older than 24 h	1. Age complaint by 25 h 2. Run escalation check 3. View as chief warden	SLA: HIGH 24 h, MEDIUM 48 h, LOW 72 h	Complaint becomes ESCALATED, assigned to chief warden, chief warden notified; new complaints untouched	API: HIGH complaint aged 25 h escalated to Dr. Lakshmi Narayanan; 0-h LOW complaint not escalated; chief warden notified	**Pass**	—
FT-14	**Student registers a visitor**	Visitors	Logged in as student	1. Fill visitor name, relation, phone, date, time 2. Submit 3. Submit duplicate	Sunil Reddy, PARENT, +3 days 10:00-12:00	Request PENDING; wardens notified; duplicate -> 409	API: 201 PENDING; 12 warden notifications; duplicate same visitor/date -> 409 UI: request shown Pending to student → warden approved with remark → student list shows Approved + 'Please carry ID'	**Pass**	—
FT-15	**Warden approves a visitor request**	Visitors	Pending request	1. Warden opens Visitors 2. Approve with remark 3. Student checks status	Remark 'Please carry ID'	Status APPROVED shown to student with remark; notification sent; second decision -> 409	API: Warden approval -> 200 APPROVED; student sees APPROVED and gets notification; deciding again -> 409 UI: request shown Pending to student → warden approved with remark → student list shows Approved + 'Please carry ID'	**Pass**	—
FT-16	**Warden rejects a visitor request**	Visitors	Pending request	1. Reject without reason 2. Reject with reason	Reason 'Friends only on weekends'	No reason -> 422; with reason -> REJECTED and remark shown to student	API: Reject without reason -> 422 'Please give a reason when rejecting a visitor request'; with reason -> REJECTED, remark shown to student	**Pass**	—
FT-17	**Invalid visitor date and time**	Visitors	Logged in as student	1. Past date 2. Date > 30 days ahead 3. Malformed date 4. End before start 5. Outside 08:00-20:00	2020-01-01, +45 days, 15:00-14:00	Each rejected with a clear message (422 / inline error)	API: past date -> 422 'Visit date cannot be in the past'; more than 30 days ahead -> 422 'Visits can be requested at most 30 days in advance'; malformed date -> 422 'Visit date must be a valid date (YYYY-MM-DD)'; end before start -> 422 'Visit end time must be after the start time'; outside visiting hours -> 422 'Visiting hours are 08:00 to 20:00'; missing phone -> 422 'visitor_phone: Phone must be a valid 10-digit Indian mobile ' UI: 'Visit date cannot be in the past' shown; no request created	**Pass**	—

TC ID	Test Case Description	Module	Pre-Conditions	Test Steps	Test Data (Input)	Expected Result	Actual Result	Status	Defect ID
FT-18	**Sick-meal request and fulfilment**	Hostel services	Student with room; mess staff	1. Request DINNER sick meal 2. Mess staff accepts 3. Marks completed	Diet note 'Soft food'	PENDING -> ACCEPTED -> COMPLETED; only mess staff can act	API: Sick meal 201 PENDING -> in mess queue -> ACCEPTED -> COMPLETED by mess staff; cleaning staff -> 403	**Pass**	—
FT-19	**Room-cleaning request**	Hostel services	Student with room	1. Request MORNING cleaning 2. Duplicate 3. Missing slot 4. Date > 7 days	Tomorrow, MORNING	Created and visible to cleaning staff; invalid requests rejected	API: Cleaning 201 PENDING visible to cleaning staff; duplicate -> 409; missing slot -> 422; date > 7 days -> 422	**Pass**	—
FT-20	**Raise, acknowledge and resolve an SOS**	Emergency SOS	Student, warden and security accounts	1. Click SOS 2. Confirm in dialog 3. Warden acknowledges 4. Resolves with note	'Twisted ankle on stairs'	Alert stored with timestamps; warden + security notified < 5 s; RAISED -> ACKNOWLEDGED -> RESOLVED	API: SOS RAISED, 6 responders (warden, chief, security) notified in 2.2 ms (API round trip 15 ms); resolve-before-ack -> 409; ACKNOWLEDGED -> RESOLVED with timestamps UI: confirmation dialog with demo disclaimer → alert Raised on warden board → Acknowledged → Resolved; student sees Resolved	**Pass**	—
FT-21	**SOS without confirmation / by unauthorised role**	Emergency SOS	Logged in	1. Send SOS with confirm=false 2. Maintenance staff tries to raise / acknowledge	confirm=false	400 and no alert stored; staff without security role -> 403	API: Unconfirmed SOS -> 400 'SOS must be confirmed before it is sent', no alert stored; maintenance staff cannot acknowledge (403)	**Pass**	—
FT-22	**Create an inter-hostel delivery request (₹10)**	P2P delivery	Sender and receiver in different hostels	1. Send an item 2. Enter receiver reg. no. + item 3. Confirm ₹10 fee 4. Try same-hostel receiver	Receiver 24BCE1001	Request OPEN, fee ₹10, demo receipt issued; same hostel / unconfirmed fee / unknown receiver rejected	API: 201 DLV-00002 OPEN, fee ₹10, demo receipt RCPT-DEMO-000002, Men's Hostel North -> Ladies Hostel East; same-hostel receiver -> 400; fee not confirmed -> 400; unknown receiver -> 404 UI: DLV-00002 posted with ₹10 demo fee → accepted & picked up by carrier → receiver saw 6-digit OTP and QR → wrong OTP showed '2 attempt(s) left' → correct OTP → Delivered (also on sender's view)	**Pass**	—
FT-23	**Carrier accepts a delivery**	P2P delivery	Open delivery request	1. Another student opens 'Available to carry' 2. Accepts 3. Sender/receiver/second carrier try to accept	Carrier Vikram / Arjun	ACCEPTED with carrier recorded; sender/receiver -> 400; second carrier -> 409	API: Listed as available; sender/receiver accept -> 400; Vikram accepts -> ACCEPTED; second carrier -> 409 'This delivery has already been taken by another carrier' UI: DLV-00002 posted with ₹10 demo fee → accepted & picked up by carrier → receiver saw 6-digit OTP and QR → wrong OTP showed '2 attempt(s) left' → correct OTP → Delivered (also on sender's view)	**Pass**	—
FT-24	**Handover with valid OTP**	P2P delivery	Delivery picked up	1. Carrier marks picked up 2. Receiver reads OTP 3. Carrier enters OTP	6-digit OTP from receiver	OTP only visible to receiver; status DELIVERED; OTP cleared	API: OTP visible only to receiver (hidden from carrier); carrier entered it -> 200 DELIVERED; OTP cleared after use UI: DLV-00002 posted with ₹10 demo fee → accepted & picked up by carrier → receiver saw 6-digit OTP and QR → wrong OTP showed '2 attempt(s) left' → correct OTP → Delivered (also on sender's view)	**Pass**	—
FT-25	**Handover with invalid OTP**	P2P delivery	Delivery picked up	1. Carrier enters wrong OTP 2. Non-numeric OTP 3. Sender enters OTP	000000, '12ab'	400 with attempts left; 422; 403; status stays PICKED_UP	API: Wrong OTP -> 400 'Incorrect OTP. 2 attempt(s) left.'; non-numeric -> 422; sender entering OTP -> 403; status stays PICKED_UP	**Pass**	—
FT-26	**OTP lock after 3 attempts and expiry**	P2P delivery	Delivery picked up	1. Enter wrong OTP 3 times 2. Enter correct OTP 3. Re-issue OTP and let it expire 4. Re-issue limit	3 x 000000; expiry 10 min	Locked (423) after 3rd wrong try; expired -> 410; at most 3 OTPs per delivery	API: 3 wrong OTPs -> 400, 400, 423 (locked); correct OTP after lock -> 410; expired OTP -> 410 'OTP has expired or is not active. Ask for a new OTP.'; OTP re-issue capped at 3 (4th -> 409)	**Pass**	—
FT-27	**Invalid delivery state transitions**	P2P delivery	Open / accepted delivery	1. Verify OTP before pickup 2. Pickup by non-carrier 3. Cancel after acceptance 4. Fail before pickup	-	403 / 409 responses; state unchanged	API: Non-carrier verify/pickup -> 403; cancel after ACCEPTED -> 409 'Invalid delivery state change from ACCEPTED to CANCELLED'; fail before pickup -> 409	**Pass**	—
FT-28	**Handover with QR code + photo + receiver details**	P2P delivery	Delivery picked up	1. Carrier scans receiver's QR 2. Uploads parcel photo 3. Enters name, reg. no., room, block	Kavya Reddy, 24BIT1005, 203, H	Wrong details / forged QR / fake image rejected; correct data -> DELIVERED with photo stored	API: Wrong room -> 400 'Verification failed: room number did not match'; forged QR -> 400; fake image -> 415; correct QR + photo + details -> DELIVERED with photo stored	**Pass**	—

TC ID	Test Case Description	Module	Pre-Conditions	Test Steps	Test Data (Input)	Expected Result	Actual Result	Status	Defect ID
FT-29	**Parcel reservation with courier name and gate OTP**	Gate delivery	Student; security staff	1. Book parcel with courier name 2. Security marks arrived 3. Verifies student's OTP	BlueDart, courier Ravi Kumar	BOOKED -> AT_GATE -> COLLECTED; missing courier name, wrong OTP and non-security staff rejected	API: Missing courier name -> 422; GATE-00001 BOOKED with courier 'Ravi Kumar'; collect before arrival -> 409; security marks AT_GATE; wrong OTP -> 400; maintenance staff -> 403; correct OTP -> COLLECTED	**Pass**	—
FT-30	**Warden publishes a notice**	Notices	Warden account	1. Publish notice to Ladies Hostel East 2. Student of that hostel views board 3. Student of other hostel views	'Fire drill on Friday' (URGENT)	Notice visible to target hostel only; students notified; empty notice -> 422	API: Warden published URGENT notice to Ladies Hostel East; shown to Priya, hidden from Men's hostel student; 15 students notified; empty notice -> 422 UI: notice published by warden; visible on student's notice board and in student's notifications	**Pass**	DEF-07 (fixed)
FT-31	**Unauthorised administrative actions by staff/wardens**	Access control	Warden, security, maintenance accounts	1. Warden deactivates user / creates hostel 2. Security approves visitor 3. Maintenance views audit log	-	403 for every attempt	API: 403 for: warden deactivates user; warden creates hostel; security approves visitor; maintenance views audit log	**Pass**	—
FT-32	**Data persists after page reload**	Persistence	Complaint just created	1. Create complaint in UI 2. Reload the page	-	Complaint still listed with the same code	UI: CMP-00008 created (status Open, forwarded to maintenance staff) and still listed after page reload	**Pass**	—
FT-33	**Report and search lost & found items**	Community	Logged in	1. Report lost umbrella 2. Search 'umbrella' 3. Another student closes it 4. Reporter closes it	Black umbrella, R block lobby	Item found by search; only reporter/warden can mark returned; future date rejected	API: Lost item reported (201) and found by keyword search; future date -> 422; only reporter can close (others 403)	**Pass**	—
FT-34	**Marketplace listing and sale**	Community	Student accounts	1. List item 2. Negative price 3. Non-seller marks sold 4. Seller marks sold 5. Sort by price	Drafter set ₹250	Listing created; invalid / unauthorised changes rejected; sold items hidden; sorting correct	API: Listing 201; negative price -> 422; non-seller mark sold -> 403; sold item hidden from listings; price sort ascending verified	**Pass**	—
FT-35	**Notes sharing and private document vault**	Community	Student accounts	1. Share note with PDF 2. Upload vault document 3. Other student tries to download 4. Upload .exe / > 2 MB	allot.pdf, x.exe, 2 MB+	Notes readable by classmates; vault file only for owner (404 for others); bad files rejected	API: Note with PDF shared and readable by classmates; vault file downloadable by owner, other student -> 404; .exe -> 415; >2 MB -> 413	**Pass**	—
FT-36	**Mess feedback**	Community	Student account	1. Rate breakfast 2. Rate same meal again 3. Rating out of range	Food 4, service 5; rating 7	Saved; duplicate -> 409; invalid -> 422; mess staff sees averages	API: Feedback 201; same meal twice -> 409; rating 7 -> 422; mess staff summary shows BREAKFAST avg 3.0	**Pass**	—
FT-37	**Roommate finder**	Accommodation	Students with preferences	1. Open roommate finder 2. Warden calls the same API	Priya's preferences	Same-gender matches ranked by preference score; rooms with free beds; warden -> 403	API: 9 same-gender matches ranked by preference score (top Sneha Das 3/3); 46 rooms with free beds listed	**Pass**	—
FT-38	**Room allocation rules**	Accommodation	Admin; student without room	1. Allocate female student to boys' hostel 2. To ladies' room 3. To full room 4. Student allocates 5. Duplicate room no.	Room G-xxx	Wrong hostel 400; valid 200; full 409; student 403; duplicate room 409	API: Female student -> boys' hostel 400; -> G-104 200; full room -> 409; student allocating -> 403; duplicate room number -> 409	**Pass**	—
FT-39	**Admin reports match stored data and export CSV**	Reports	Admin account	1. Open reports 2. Compare totals with database 3. Export complaints CSV 4. Warden opens reports	-	Report totals equal DB counts; CSV rows = complaints; warden -> 403	API: Report complaint total 7 = DB count; students 57 = DB count; CSV export has 7 rows + header; warden -> 403	**Pass**	—
FT-40	**In-app messaging between student and warden**	Messaging	Student and warden	1. Student sends message 2. Warden opens conversation 3. Empty message	'Can my parents visit on Sunday?'	Message delivered with unread count, marked read when opened; empty -> 422	API: Student -> warden message delivered (unread=1), shown in thread, marked read on open (unread=0); empty message -> 422	**Pass**	—
FT-41	**Logout invalidates the session**	Authentication	Logged-in user	1. Log out 2. Reuse old token / open protected page	-	Token rejected with 401; UI redirects to login	API: Logout 200; reusing the same token -> 401 'Session has been logged out' UI: logout returns to /login; opening /complaints afterwards redirects to /login	**Pass**	—
FT-42	**Administrator registers warden, staff and admin accounts**	Registration	Logged in as admin	1. Create WARDEN, STAFF (security) and ADMIN users 2. Each logs in 3. Staff without type 4. Duplicate email	new.warden@vista.demo / Strong123	Accounts created with the right role and can log in; invalid / duplicate rejected	API: Admin created WARDEN, STAFF (SECURITY) and ADMIN accounts (201) and each could log in; staff without type -> 422; duplicate email -> 409	**Pass**	—

TC ID	Test Case Description	Module	Pre-Conditions	Test Steps	Test Data (Input)	Expected Result	Actual Result	Status	Defect ID
FT-43	**Hostel map and hostel structure**	Accommodation	Logged in	1. Open Hostel Map 2. Check blocks, mess, gates, service points 3. Check hostel occupancy 4. Call without login	-	All location types shown; hostels list blocks and occupancy; unauthenticated -> 401	API: Map returns 12 locations (BLOCK, GATE, MESS, SERVICE); 4 hostels with blocks and live occupancy; unauthenticated -> 401	Pass	—

Functional Testing Summary

Total Test Cases	Executed	Passed	Failed	Blocked	Not Run	Pass Percentage
43	43	43	0	0	0	100%

Pass percentage = Passed ÷ Executed × 100. Executed = test cases with a Pass or Fail outcome; Blocked and Not Run are reported separately.

Key Takeaways

- ✔ 43 of 43 functional test cases were executed against the running application; 43 passed and 0 failed (pass rate 100%).
- ✔ Evidence comes from 61 API tests (pytest) and 23 browser tests (Playwright, Chromium) that drive the real React UI, FastAPI backend and PostgreSQL database.
- ✔ The first execution failed 6 browser tests and 1 API test because of test-script problems (DEF-07) – fixed and re-run; no functional requirement failed after the fixes.
- ✔ A real data-integrity defect (DEF-01, duplicate complaints on parallel submit) was found by testing and fixed.
- ✔ The unique P2P delivery flow (₹10 fee, OTP lock after 3 tries, OTP expiry, QR + photo check) and SOS alerts were verified end to end.
- ✔ Payments are demo records only and SOS is not connected to real emergency services.

Results by module

Module	Test cases	Passed	Failed	Blocked / Not Run	Case IDs
Authentication	4	4	0	0	FT-01, FT-02, FT-03, FT-41
Access control	2	2	0	0	FT-04, FT-31
Registration	3	3	0	0	FT-05, FT-06, FT-42
Profile	1	1	0	0	FT-07
Complaints	6	6	0	0	FT-08, FT-09, FT-10, FT-11, FT-12, FT-13
Visitors	4	4	0	0	FT-14, FT-15, FT-16, FT-17
Hostel services	2	2	0	0	FT-18, FT-19
Emergency SOS	2	2	0	0	FT-20, FT-21
P2P delivery	7	7	0	0	FT-22, FT-23, FT-24, FT-25, FT-26, FT-27, FT-28
Gate delivery	1	1	0	0	FT-29
Notices	1	1	0	0	FT-30
Persistence	1	1	0	0	FT-32
Community	4	4	0	0	FT-33, FT-34, FT-35, FT-36
Accommodation	3	3	0	0	FT-37, FT-38, FT-43
Reports	1	1	0	0	FT-39
Messaging	1	1	0	0	FT-40